Green and Sustainable Pharmaceutical Production

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Green and Sustainable Pharmaceutical Production (A34162)

Short Title:	Green Sus Pharm Prodn		
Faculty	Science and Computing		
Department	Science		
Cluster	Environmental Science		
Author	CLENNON		
Credits	10	**Level:**	Postgraduate

Description of Module / Aims

This module will provide students with a detailed understanding of the legislative, environmental, social and economic effects of climate change and the response required by the pharmaceutical industry. Students will gain clear insight into green and sustainable principles, tools and methodologies for the synthesis, analysis, formulation and packaging of pharmaceuticals, including the application of quality management principles and renewable energy technologies. Students will critically assess case study examples and process changes and will develop key skills to deploy current state of the art in this area within an organisation.

Programmes

PHAR-0020	Certificate in Green and Sustainable Pharmaceutical Production (WD_SGSPP_MA)	5 / 0 / M
PHAR-0020	MSc in Analytical Science with Quality Management (WD_SANSC_R)	1 / 0 / E
PHAR-0020	PG Dip in Analytical Science with Quality Management (WD_SANSC_G)	1 / 0 / E

Indicative Content

- Sustainability, UN Sustainable Development Goals, climate legislation and targets, circular economy, social and economic impact of climate change.
- Sustainable drug discovery, development and processing. Green Chemistry and Engineering Principles, Metrics for measurement of process change impact, life cycle assessment.
- Process feedstock, solvent and reagent selection tools, biomass based alternatives to petroleum-based chemicals, alternative solvents such as DES and IL, solvent free processes, reuse and recycling of materials.
- Toxicological assessment of materials and products, LD 50, OEB and OEL.
- Sustainable API synthesis methodology examples such as catalysis-organo- and biocatalysis, flow chemistry, photochemistry, ultrasound and microwaves.
- Sustainable Analytical methodology, sampling protocols, chromatography, method development, PAT.
- Sustainable formulation methodologies including processing methods, green drug delivery platforms, packaging and stability.
- Lean tools applied to sustainability, green lean six sigma for sustainable development.
- Integration of renewable electrical and heat systems into pharmaceutical and fine chemical plants.
- Feasibility assessment, financial appraisal

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate the legislative, environmental, social and economic aspects associated with climate change and sustainability.
2. Appraise green and sustainable principles and tools for the synthesis, analysis and formulation, of pharmaceuticals and fine chemicals, including quality management principles.
3. Assess the challenges and opportunities for sustainable packaging for Pharmaceutical/ Biopharmaceutical/Medtech products.
4. Critically interpret the application of Green and Sustainable principles through key case studies and examples.
5. Critique the impact of green and sustainable process changes applied to a pharmaceutical or fine chemical process.
6. Assess candidate sites for renewable energy deployments using the latest energy technologies.
7. Develop a strategy to employ green and sustainable principles within an organisation incorporating current state of the art with relevant literature examples.

Learning and Teaching Methods

- Blended learning with synchronous and asynchronous activities of lectures, project and presentations

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4,5,6,7
Participation	20%	2,3,4,5,6
In-Class Assessment	30%	1,2,3,4,6
Project	30%	7
Presentation	20%	7

Assessment Criteria

<40%: The learner will demonstrate little or no understanding of the fundamentals of the learning outcomes. Unable to apply the knowledge in an industrial context.

40%-59%: The learner will demonstrate limited understanding of the fundamentals of the learning outcomes. Some attempt will be made at application of theory to practice.

60%-69%: The learner will demonstrate good understanding of the fundamentals of the learning outcomes, with critical evaluation and individuality of thought.

70%-100%: Very good level of critical analysis, originality of thought and comprehensive knowledge of the subject area. Able to evaluate and implement change.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		48
Independent Learning		222

Essential Material

- De Le Guardia, M. and S. Garrigues. _Handbook of Green Analytical Chemistry_. UK: John Wiley & Sons, 2012.
- Zhang , W. and B.W. Cue. _Green Techniques for Organic Synthesis and Medicinal Chemistry_. 2nd. UK: John Wiley & Sons Inc, 2018.

Supplementary Material

- "Green Chemistry - American Chemical Society (acs.org)." Green Chemistry - American Chemical Society (acs.org). https://www.acs.org/content/acs/en/greenchemistry.html
- "https://library.wit.ie/databases/." i2i National Standards Authority of Ireland NSAI (National Standards Authority of Ireland) Ireland's official standards body. https://library.wit.ie/databases/
- Anastas, P. and J. Warner. _Green Chemistry: Theory and Practice_. New York: Oxford University Press, 1998.
- Dunn, P.J., A.S. Wells and M.T. Williams. _Green Chemistry in the Pharmaceutical Industry_. Weinheim, Germany : Wiley, 2010.
- Erlich, R. and H.A. Geller. _Renewable Energy: A First Course_. UK: Routledge, 2018.
- Kumar, V., J.A. Garza-Reyes and F. Chen. _Seeing Green:Achieving Environmental Sustainability through Lean and Six Sigma_ . England, UK: Emerald Publishing Limited, 2017.
- Tao, J.T. and R.J. Kazlauskas. _...Biocatalysis for Green Chemistry and Chemical Process Development_. UK: Wiley and sons, 2011.
- Vacarro, L. _Sustainable Flow Chemistry, Methods and Applications_. UK: Wiley and Sons, 2017.